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1 Name your Jupyter Notebook as:  
TASK1_<your class>_<your name>.ipynb
```

The task is to implement a Vigenère cipher encryption algorithm.

The Vigenère cipher is a method of encrypting alphabetic text by using a simple form of polyalphabetic substitution. A key is repeated to match the length of the plaintext, and each letter in the plaintext is shifted according to the corresponding letter in the key.

For example:

The plain text is "HELLO WORLD" and the key is "ABLE".

The key, repeated to match the 11 characters in the text including spaces, is "ABLEABLEABL".

Each letter in both the plain text and key is first converted to a number (A=0, B=1, C=2, D=3, E=4, F=5, ..., Z=25). This is to be done with the help of ASCII values.

The two numbers are then added up to give the encrypted letter.

First letter 'H' in the plain text is number 7 and first letter 'A' in the key is number 0. Calculate $7 + 0 = 7$. Hence, the first encrypted letter is 'H'.

Second letter 'E' in the plain text is number 4 and second letter 'B' in the key is number 1. Calculate $4 + 1 = 5$. Hence, the second encrypted letter is 'F'.

When a letter goes beyond 'Z', it returns to 'A'. Use modular arithmetic with modulus 26 to wrap around if necessary.

Space is replaced with the character '!'.

Plaintext character	H	E	L	L	O		W	O	R	L	D
Key character	A	B	L	E	A	B	L	E	A	B	L
Encrypted character	H	F	W	P	O	!	H	S	R	M	O

The resulting encrypted text is "HFWPO!HSRMO".

Task 1.1

Write a function CharToNum() that takes an uppercase letter as parameter and returns an integer corresponding to its position in the alphabet, where 'A'=0, 'B'=1, 'C'=2, ..., 'Z'=25. You are required to use ASCII encoding in your implementation.

[2]

Task 1.2

Write a function encrypt() that takes a string text and a string key as input parameters.

The encrypt() function returns an encrypted string where each letter in text is shifted forward by the corresponding letter in key. You are required to use the function from Task 1.1 in your implementation.

Assume both the text and key contain only uppercase letters, and spaces are allowed in the text. No other characters are in the text and key.

[7]

[JPJC/2024/Q1 Cont']

Task 1.3

The text file `TASK1.txt` contains messages that needs to be encrypted. Thereafter, the encrypted messages are stored in a text file named `ENCRYPTED.txt`.

Write the program code to:

- Read the data from the text file `TASK1.txt`
- The key is "JPJC"
- Use your function from **Task 1.2** to encrypt the content
- Write the encrypted messages to the text file `ENCRYPTED.txt`

[3]

Test your program with the plain text in the file `TASK1.txt`.

Display the content of `ENCRYPTED.txt` after you have run the program.

[1]

Save your Jupyter Notebook for Task 1.